

Imaris Registration Tool

User Guide & Laboratory Documentation

ClearMap Adjustments Project

3D microscopy image registration using Imaris (.ims) data and Elastix

Current release	v0.0.1
Repository	HSHuangLab/ClearMap_Adjustments
Platform	Python / PySide6; Windows setup documented
Status	Research tool - initial laboratory release

Purpose: enable laboratory users to load microscopy volumes, configure a registration pipeline, run Elastix, and review quantitative alignment metrics through a graphical interface.

1. Overview

The Imaris Registration Tool is a graphical application developed within the ClearMap Adjustments project. It provides a practical workflow for 3D microscopy image registration using Imaris (.ims) files and Elastix.

The current version supports loading microscopy volumes, choosing a resolution level and channel, configuring one or more registration transformations, selecting an output folder, running Elastix, and evaluating the result using MAE, MSE, and NCC.

2. Main Features

- Load Imaris (.ims) microscopy files.
- Select image resolution level and channel.
- Use Translation, Rigid, and Affine transformations.
- Create multi-step registration sequences.
- Run Elastix from the graphical application.
- Keep the graphical interface responsive while registration runs in a background thread.
- Evaluate alignment using MAE, MSE, and NCC.
- Automatically save registration inputs and results to a timestamped output folder.

3. Important Notes Before Installation

This is an initial research version of the tool. It should be tested on the intended workstation and dataset before routine use. Large .ims volumes may require a lower resolution level because loading the complete volume can exceed available memory.

Elastix is an external component and is not installed by requirements.txt. The correct Elastix installation/executable for the target operating system must be available to the application.

4. Downloading the Project

Open Command Prompt, PowerShell, or the VS Code terminal and clone the repository:

```
git clone https://github.com/HSHuangLab/ClearMap_Adjustments.git
```

Enter the repository:

```
cd ClearMap_Adjustments
```

To inspect available released versions:

```
git tag
```

To use a specific release, for example v0.0.1:

```
git checkout v0.0.1
```

5. Python Environment Setup on Windows

From the ClearMap_Adjustments directory, create a virtual environment:

```
py -m venv .venv
```

5.1 Command Prompt (CMD)

Activate the virtual environment:

```
.venv\Scripts\activate
```

Install the Python dependencies:

```
python -m pip install -r requirements.txt
```

5.2 PowerShell / VS Code

On some laboratory computers, PowerShell may display "running scripts is disabled on this system" when trying to activate the environment. Changing the computer's execution policy is not required.

Instead, use the Python executable inside the virtual environment directly:

```
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
```

The application can also be launched directly with:

```
.\.venv\Scripts\python.exe .\ImarisRegistrationTool\app.py
```

6. Elastix Installation on Windows

Elastix is the registration engine used by the application. It is a native external program and is not installed by Python or by requirements.txt. A Windows-compatible Elastix distribution must therefore be installed separately.

The official Elastix project provides precompiled 64-bit Windows binaries. For a new workstation, use the Windows package from the official SuperElastix/elastix GitHub Releases page. At the time this guide was prepared, the latest official release was Elastix 5.3.1.

Official release page:

```
https://github.com/SuperElastix/elastix/releases
```

6.1 Download and Extract Elastix

1. Open the official Elastix Releases page.
2. Download the 64-bit Windows archive for the required Elastix version (for example, elastix-5.3.1-windows.zip).
3. Extract the archive to a permanent location. Do not run Elastix directly from inside the ZIP file.
4. Keep the extracted Elastix files together. The executable may depend on libraries distributed in the same package.

6.2 Verify the Installation

Locate elastix.exe in the extracted package. Open Command Prompt or PowerShell in the directory containing the executable and run:

```
.\elastix.exe --help
```

If Elastix starts and prints its command-line help, the executable is available. If Windows reports a missing runtime or DLL, install/update the Microsoft Visual C++ Redistributable for x64 and try again. The official Elastix Windows release notes specifically note that this runtime may be required.

Microsoft Visual C++ Redistributable:

<https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist>

6.3 Connect Elastix to the Registration Tool

The application must know the location of the Elastix executable. The current research version resolves Elastix from the project configuration/code. Before running the application on a new Windows workstation, verify that the configured path points to the Windows executable (elastix.exe), not to a macOS or Linux binary.

Important: copying a macOS Elastix executable to Windows will not work. Each operating system requires its corresponding native Elastix binary.

6.4 Quick Check Before First Registration

Before processing microscopy data, confirm all three items: the Python virtual environment is working, the application opens successfully, and the Windows Elastix executable can be launched. This separates Python setup problems from Elastix installation problems and makes troubleshooting much easier.

7. VS Code Configuration

Open the ClearMap_Adjustments folder in Visual Studio Code. Press Ctrl + Shift + P, select "Python: Select Interpreter", and choose:

```
.env\Scripts\python.exe
```

To check which Python is visible in Command Prompt:

```
where python
```

In PowerShell:

```
Get-Command python
```

To verify the exact Python executable used by Python:

```
python -c "import sys; print(sys.executable)"
```

When the virtual environment is configured correctly, the displayed path should point to ClearMap_Adjustments\.env\Scripts\python.exe.

8. Running the Application

From the repository root on Windows:

```
.\.venv\Scripts\python.exe .\ImarisRegistrationTool\app.py
```

The graphical application should open. Registration runs in a worker thread so the interface can remain responsive while Elastix is processing.

9. Registration Workflow

Step	Action	Description
1	Select the fixed image	Use Browse under IMAGE 1 - FIXED and select an Imaris (.ims) file.
2	Choose resolution and channel	Select the required resolution level and channel, then click Load Image.
3	Select the moving image	Repeat the same process under IMAGE 2 - MOVING.
4	Choose transformations	Select the registration sequence. Step 1 is required; Steps 2 and 3 may be set to None.
5	Choose output folder	Use Browse under Output folder and select where the registration results should be saved.
6	Run registration	Click Run Registration. The progress indicator remains visible while Elastix is running.
7	Review results	When registration completes, the application displays MAE, MSE, and NCC before and after alignment.
8	Locate saved output	The completion message displays the path to the timestamped output directory.

10. Registration Options

Translation

Corrects displacement between images without changing image orientation, scale, or shape.

Rigid

Corrects translation and rotation while preserving image size and shape.

Affine

Supports translation, rotation, scaling, and shearing.

Transformations may be applied sequentially. For example, Rigid -> Affine can first establish a simpler initial alignment before applying the more flexible affine model.

11. Alignment Metrics

Metric	Meaning	General interpretation
MAE	Mean Absolute Error	Lower generally indicates greater similarity.
MSE	Mean Squared Error	Lower generally indicates greater similarity; large differences are penalized strongly.
NCC	Normalized Cross-Correlation	Higher generally indicates greater similarity/correlation.

Important: quantitative metrics should be interpreted together with visual inspection. Registration may create invalid boundary regions when transformed content moves outside the original field of view. These regions can make full-volume metrics appear worse even when alignment in the valid overlapping region improves.

12. Output Files

Each registration run creates a new timestamped output directory. The workflow writes the fixed and moving volumes used for registration and stores the files generated by Elastix, including the final registered volume.

Using a separate timestamped directory prevents a new run from overwriting previous registration results.

13. Troubleshooting

Problem	Recommended action
"running scripts is disabled on this system"	Do not change the laboratory computer execution policy. Run <code>.venv\Scripts\python.exe</code> directly as shown in Section 5.2.
Wrong Python interpreter in VS Code	Use <code>Ctrl + Shift + P</code> -> "Python: Select Interpreter" -> <code>.venv\Scripts\python.exe</code> .
Python package/import error	Confirm dependencies were installed with the <code>.venv</code> Python and <code>requirements.txt</code> .
Image is too large	Select a lower Imaris resolution level and load the image again.
Registration does not start	Confirm both fixed and moving images are loaded, an output directory is selected, and at least one transform is configured.
Elastix cannot be found or launched	Verify that the correct Elastix executable for the workstation is present and that the application path configuration points to it.

14. Current Limitations and Future Development

- The current tool is intended for research use and has not been validated for clinical use.
- Registration performance and suitable transformation sequences depend on the dataset.
- Large datasets are constrained by available system memory.
- Additional Elastix parameters, transformation models, metrics, and optimization strategies can be investigated in future versions.
- Future documentation may include experimentally tested recommendations for specific microscopy datasets.

15. Suggested Validation Before Routine Laboratory Use

Before adopting a new configuration, inspect the registered volume visually and compare quantitative metrics. When possible, test the workflow first on data with a known or controlled transformation so that the expected registration behavior can be verified.

16. Project Files

The registration application is located under `ImarisRegistrationTool`. The project also contains the registration parameter files and supporting modules for loading Imaris data, running Elastix, and evaluating alignment.

The GitHub repository should remain the development source, while tagged releases can be used to preserve stable versions of the laboratory tool.

End of User Guide